Bayesian Statistics
Project A4

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1 Introduction

1.1 Project overview

Air pollution is a major environmental and public health issue around the world. Among the various atmospheric pollutants, tropospheric ozone plays a particularly important role due to its harmful effects on human health, vegetation, and ecosystems. High ozone concentrations are associated with respiratory diseases, reduced lung function, and increased mortality during pollution peaks. Hence, regulatory agencies therefore define thresholds beyond which public information and alert procedures must be triggered in order to protect vulnerable populations.

Despite emission control programs implemented in many urban areas, ambient ozone concentrations still frequently exceed recommended limits. This persistence highlights the importance of developing reliable tools to understand, monitor, and predict high ozone episodes. In particular, extreme ozone events drive most public health warnings and environmental policies. Accurately modeling these extremes is a crucial task for environmental monitoring and risk management.

Traditionally, air pollution modeling approaches are divided into deterministic and statistical methods. Deterministic models attempt to solve the equations governing pollutant transport and transformation. However, those models can be insufficient depending on the nature of the data. Statistical approaches model the relationship between ozone and explanatory variables such as meteorology and seasonality. These methods are often more flexible and allow to directly compute and predict extreme quantiles for a given day.

This project focuses on the statistical modeling of maximum ozone concentrations using meteorological covariates. Extreme value theory (EVT) is particularly suited for analyzing high pollution episodes, which are most relevant from a regulatory and public health perspective. Since extreme ozone events drive policy decisions and alert procedures, modeling the tail behavior of the distribution is essential for reliable risk assessment. The proposed approach aims to separate underlying trends and seasonality from meteorological effects, while allowing direct inference on rare but critical events. Nevertheless, classical extreme value methods present challenges, such as non-stationarity, the difficulty of defining independent clusters, and the loss of information when non-peak observations are discarded.

1.2 Objectives and methodological challenges

The main scientific objective of this project is to understand how meteorological conditions influence extreme ozone concentrations and to develop a bayesian statistical model capable of capturing this relationship in a reliable and interpretable way. More precisely, the focus is on daily maximum ozone levels observed at different monitoring stations and on modeling these extremes using the Generalized Extreme Value (GEV) distribution, which is standard in the analysis of environmental extremes. We fit a model for the extreme values of ozone, denoted by Y_{ij}, corresponding to day i at station j.

As a first step, the relationship between ozone maxima and several meteorological variables is explored. This exploratory analysis helps identify the dominant drivers of ozone variability and highlights the most influential predictors associated with extreme pollution episodes.

The project relies on extreme value theory within a Bayesian framework. Modeling extremes, however, raises methodological challenges, as classical regression assumptions of finite moments may not hold. In particular, certain parameter values of extreme value distributions can imply infinite mean or variance, making standard location–scale parameterizations difficult to interpret and numerically unstable. To address this issue, we adopt a reparameterization based on quantiles and quantile ranges, which remain well-defined even when moments do not exist, thereby improving interpretability and computational stability.

Another important challenge concerns the specification of prior distributions. Inadequate priors may lead to unrealistic inference or numerical instability. To mitigate this risk, we adopt Penalized Complexity (PC) priors, which shrink the model toward a simpler base structure unless the data support additional complexity. We also introduce a truncated and renormalized version, the P3C prior, which restricts the parameter space to ensure finite mean and variance, thereby improving interpretability and robustness.

1.3 Dataset description

We worked with two datasets to obtain information on ozone concentration and related meteorological covariates. The Arpa Lombardia datsets provides hourly measurements of several air pollutants, including ozone, from a network of fixed monitoring stations across the region. The measurements cover the period from the 01/01/2018 to 31/12/2024 (2500 days per station), allowing the analysis to capture both short-term variability and longer-term patterns. The final datasets includes 51 monitoring stations.
The open-meteo daraset is an open-source API offering meteorological variables at specific locations and time resolutions.
From these sources, we computed the daily maximum ozone concentration for each station and collected the following daily meteorological variables:

- Maximum temperature at 2m (°C)
- Minimum temperature at 2m (°C)
- Total daily precipitation (mm)
- Maximum wind speed at 10m (km/h)

1.4 Structure of the report

The report is organized as follows. First, an exploratory data analysis is presented to describe the dataset and the relationships between ozone and meteorological variables. Next, the statistical model and its reparameterization are introduced, followed by the specification of PC and P3C priors. The inference procedure and computational aspects are then detailed, in particular with INLA implementation. Finally, the results are presented with interpretations, with a look also to the STAN implementation of the model.

2 Data analysis

The final dataset contains 51 monitoring stations across Lombardia. For each station, daily ozone and meteorological measurements are collected (2500 days per station).
Before modeling, the data undergo preprocessing steps such as handling missing values, checking for outliers, and ensuring consistency across variables. We decided to remove days with more than 70% missing ozone measurements in a station. Moreover, some stations stopped recording before 2024. In total, the dataset contains 127,000 daily ozone observations with their associated covariates.
Exploratory analysis is performed to visualize distributions and examine pairwise relationships between ozone and meteorological predictors

2.1 Temporal aspects

The analysis of the temporal evolution of daily maximum ozone concentrations reveals a clear annual seasonality, with higher levels observed during the summer months. Moreover, no evident long-term trend is detected over the observation period. This suggests that, while ozone concentrations vary predictably throughout the year, there has been no systematic increase or decrease in extreme levels over time. (Figure A.1)

The spatio-temporal variogram (Pebesma and Gräler, 2025) shows a strong temporal decoherence as the time lag increases but no clear spatial decoherence as the distance increases (Figure A.2). The cross-correlation analysis between monitoring stations confirms that temporal dependence dominates, with ozone levels at different stations remaining highly correlated over space. (Figure A.3)

2.2 Relationship between maxOzone and covariates

The analysis of meteorological covariates shows that temperature-related variables are the most strongly correlated with daily maximum ozone concentrations. Both maximum and minimum temperatures exhibit a positive correlation with ozone peaks, indicating that warmer conditions favor higher ozone levels. In contrast, rainfall and wind speed show weaker correlations, suggesting a smaller influence on extreme ozone events. These findings highlight the dominant role of temperature in ozone levels.(Figure A.4)

3 Model

3.1 Generalized Extreme Value (GEV) Distribution

The Generalized Extreme Value (GEV) distribution describes the limiting distribution of properly normalized block maxima (Escarela, 2012). It arises from the Extremal Types Theorem, which states that the maximum of a large sample of independent and identically distributed random variables converges in distribution to a GEV family.
The GEV distribution is parameterized by:

- a location parameter $\mu \in \mathbb{R}$,
- a scale parameter $\sigma > 0$,
- a shape parameter $\xi \in \mathbb{R}$.

Its cumulative distribution function (CDF) is given by

$$F(y) = \exp\left\{ -\max(0, 1 + \xi \left(\frac{y-\mu}{\sigma} \right))^{-1/\xi} \right\},$$

The shape parameter ξ determines the tail behaviour:

- $\boldsymbol{\xi < 0}$: **Weibull type** — distribution with finite upper bound,

$$F(x; \mu, \sigma, \xi) = \begin{cases} \exp\left(-\left(\frac{\mu - x}{\sigma} \right)^{-1/\xi} \right), & x < \mu, \\ 1, & x \geq \mu, \end{cases}$$

- $\boldsymbol{\xi = 0}$: **Gumbel type** — light-tailed distribution with unbounded support,

$$F(x; \mu, \sigma, 0) = \exp\left(-\exp\left(\frac{\mu - x}{\sigma} \right) \right),$$

- $\boldsymbol{\xi > 0}$: **Fréchet type** — heavy-tailed distribution with finite lower bound,

$$F(x; \mu, \sigma, \xi) = \begin{cases} 0, & x \leq \mu, \\ \exp\left(-\left(\frac{x - \mu}{\sigma} \right)^{-1/\xi} \right), & x > \mu. \end{cases}$$

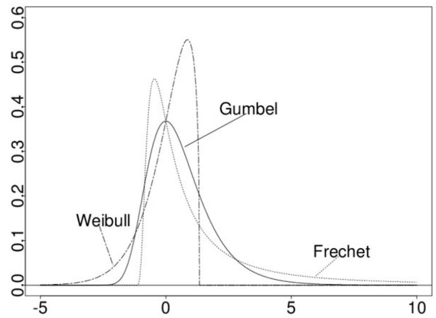


Figure 1: Illustration of the different GEV subfamilies.

Moment existence

A key limitation of the GEV distribution concerns the existence of moments. The k-th moment exists if and only if

$$\xi < \frac{1}{k}.$$

In particular:

$$\mathbb{E}[X] < \infty \iff \xi < 1, \qquad \mathrm{Var}(X) < \infty \iff \xi < \frac{1}{2}.$$

If ξ is large, the distribution may have infinite mean or variance, which complicates interpretation and prediction. This issue is particularly relevant in environmental applications where risk estimation relies on stable moment properties.

3.2 Reparametrization of the GEV

The classical parametrization (μ, σ, ξ) may lose interpretability when ξ approaches 1, since moments cease to exist. Moreover, μ and σ depend strongly on ξ, making inference unstable.
Following Castro-Camilo et al. (2022), we adopt a quantile-based parametrization:

$$q_\alpha = \text{quantile of order } \alpha, \qquad s_\beta = q_{1-\beta/2} - q_{\beta/2}.$$

Here:

- q_α represents a meaningful location measure,

- s_β represents a quantile range, describing dispersion.

These quantities remain well-defined for all $\xi \in \mathbb{R}$, even when moments do not exist. This parametrization improves interpretability and numerical stability.

$$F(y \mid q_\alpha, s_\beta) = \exp\left(\exp\left(-\frac{y - q_\alpha}{s_\beta\left(\ell_{\beta/2} - \ell_{1-\beta/2}\right) - \ell_\alpha}\right)\right).$$

3.3 Blended GEV (bGEV) Distribution

The standard GEV with $\xi > 0$ (Fréchet) implies a finite lower bound, which may be unrealistic in environmental applications such as ozone concentration modelling.
To overcome this limitation, we adopt the blended GEV (bGEV) distribution. The bGEV combines:

- the left tail of a Gumbel distribution ($\xi = 0$),
- the right tail of a Fréchet distribution ($\xi > 0$).

This construction removes the artificial lower bound while preserving heavy-tail behaviour for extremes.

$$H(x \mid q_\alpha, s_\beta, \xi) = F(x \mid q_\alpha, s_\beta, \xi)^{p(x)}\ G(x \mid \tilde{q}_\alpha, \tilde{s}_\beta)^{1-p(x)}.$$

3.4 Penalized Complexity (PC) Priors

To control tail behaviour, we introduce Penalized Complexity (PC) priors for ξ.
The base (simpler) model is:

$$\xi = 0 \quad \text{(Gumbel distribution)}.$$

The more flexible model corresponds to $\xi > 0$. The PC prior shrinks the model toward the base case and penalizes increasing deviation.
The prior density is:

$$\pi(\xi) = \lambda\sqrt{2}\exp\left(-\lambda\sqrt{\frac{2\xi}{1-\xi}}\right)\left(\frac{1-\xi/2}{(1-\xi)^{3/2}}\right), \qquad \xi \in [0, 1).$$

This prior:

- favors lighter tails unless supported by data,
- prevents overfitting of extreme behaviour.

3.5 P3C Prior: Truncated PC Prior

When $\xi > 1/2$, the variance becomes infinite. To guarantee finite mean and variance, we truncate and renormalize the PC prior:

$$\tilde{\pi}(\xi) = \frac{\pi(\xi)}{\int_0^{0.5} \pi(s)\, ds}, \qquad 0 \leq \xi < 0.5.$$

This truncated version is referred to as the **P3C prior**. It ensures:

$$\mathbb{E}[X] < \infty, \qquad \mathrm{Var}(X) < \infty.$$

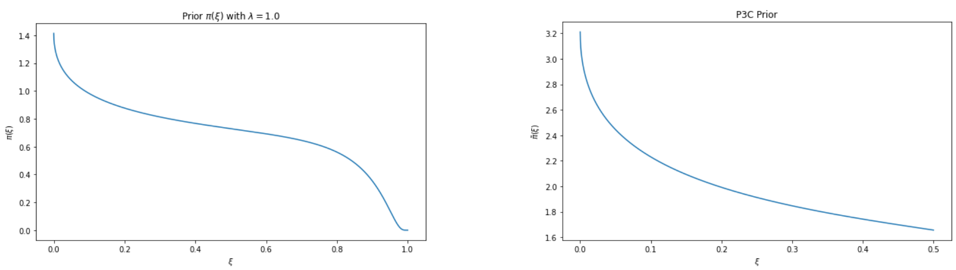


Figure 2: Comparison between PC and P3C priors.

The P3C prior is particularly important in environmental applications (e.g., ozone levels, pollution, climate extremes), where unrealistic heavy tails may lead to unstable predictions.

3.6 Bayesian Hierarchical Model

We assume daily ozone observations are conditionally independent and follow a blended GEV distribution:

$$y_{i,t} \mid q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi \overset{\text{ind}}{\sim} \text{bGEV}(q_{\alpha,i,t}, s_{\beta,i,t}, \xi),$$

for monitoring stations $i = 1, \ldots, S$ and days $t = 1, \ldots, T$.
The parameters depend on meteorological covariates:

$$q_{\alpha,i,t} = \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta}, \qquad s_{\beta,i,t} = \exp(\mathbf{x}_{i,t}^{\top}\boldsymbol{\gamma}).$$

The exponential link ensures positivity of the quantile range.
We assign independent priors:

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \Sigma_\beta), \qquad \boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \Sigma_\gamma), \qquad \xi \sim \tilde{\pi}(\xi).$$

Here:

- S is the number of monitoring stations;
- T is the number of observed days;
- $\mathbf{x}_{i,t}$ contains meteorological covariates (e.g., temperature, wind speed, humidity).

This hierarchical structure allows us to:

- capture spatial and temporal variability,
- model extreme ozone concentrations,
- ensure stable predictive behaviour through prior regularization.

4 INLA implementation and results

In order to analyze the inference procedure and the computational aspects of such a blended GEV model like the one we have just introduced, it's now time to deal with the INLA package in R.

The integrated nested Laplace approximation (INLA) is a method for approximate Bayesian inference. In recent years, it has established itself as an alternative to other methods such as Markov chain Monte Carlo because of its speed and ease of use via the R-INLA package. Although the INLA methodology focuses on models that can be expressed as latent Gaussian Markov random fields (GMRF), this encompasses a large family of models that are used in practice. The GEV model we are considering is part of this large family.

For this reason, once introduced the necessary libraries for the application of INLA package in R, we started by defining two functions that we are going to use in the next steps: the first one permits the conversion of the bGEV quantile based parametrization into the classic parameters μ (location), σ (scale) and ξ (tail); the second one, instead, is useful to map the tail parameter ξ

from the real space to a limited interval by making a logistic transformation. After the load of the dataset with the weekly ozone observations and the scale of the covariates, we opted for splitting the data into a training set, constituted by all observations from the period $2018 - 2023$ for 32 stations out of 51, and a test set, characterized by all observations of year 2024, and then decided to perform inference using the the two previously introduced functions.
In particular, we tested two different models:

- a model with a trend, a seasonal component and covariate free spread and tail;
- a second model without temporal component but with covariates in the spread.

In both cases, the aim was to fit the proposed model using the INLA framework and the bGEV likelihood, implemented using a quantile-based parametrisation centered on the location parameter $q_{\alpha,i,t}$, the spread parameter $s_{\beta,i,t}$ and the tail behaviour ξ. The central part of the distribution and the tail behaviour were smoothly blended in quantile space over the probability interval $[0.05, 0.20]$ using a Beta$(5, 5)$ mixing distribution. This construction ensured a smooth transition between the bulk and the tail component, improving numerical stability and inference.

4.1 bGEV temporal model

This section is based on the implementation of the following model with temporal components applied on the location parameter:

$$
\begin{aligned}
y_{i,t} \mid q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi &\overset{\text{ind}}{\sim} \text{bGEV}\big(q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi\big), \quad \text{for } i = 1, \ldots, S, t = 1, \ldots, T \\
q_{\alpha,i,t} &= \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta} + \beta_{year} year + f(month) \\
s_{\beta,i,t} &\sim \text{log-gamma}(3, 3) \\
\boldsymbol{\beta} &\sim \mathcal{N}\big(\mathbf{0},\, 1\big), \\
\xi &\sim \text{P3C prior}, \quad \xi \in [0,\, 0.5]
\end{aligned}
$$

In particular, from this model we can observe a linear predictor for the year and a cyclic second-order random walk (RW2) for the monthly seasonal component. This specification permits us to penalise second-order differences, enforcing smoothness in the seasonal pattern while ensuring periodicity between December and January.

Using the INLA package in R, we first specified an initial value for the tail parameter ($\xi = 0.08$), which was mapped onto the interval supported by the INLA implementation. We then constructed the corresponding PC prior, fixing the penalization parameter to $\lambda = 7$ in order to strongly discourage heavier tails compared to lighter ones. For the spread parameter, we adopted a log-Gamma prior with parameters $(3, 3)$ and combined the resulting hyperparameters within the bGEV specification. The location, spread, and blending region parameters were fixed accordingly and provided to INLA. In addition, the monthly seasonal component was modeled using a RW2 prior, with the smoothness parameter chosen to control excessive variability while preserving seasonal flexibility.

After verifying the quantities computed and predicted by INLA, we constructed the model formula, prepared the input structure, and fitted the model using the "inla" function

Summarizing the results of this fitted model (Figure A.5), we can say that:

- the intercept (108.18 μg/m^3) represents the baseline weekly median ozone concentration in the reference year (2018), when all meteorological covariates are at their centered values and the seasonal effect is at its average level;
- the year coefficient (-0.61) implies a strong decreasing trend of the median of weekly ozone observations in time;
- the minimum temperature coefficient (-2.50) tells us that higher minimum temperatures are associated with lower ozone maxima;
- the maximum temperature is the dominant driver and its coefficient (40.4) shows how hot conditions strongly enhance ozone extremes;
- the wind speed coefficient (0.56), instead, presents moderate positive effect on the median;
- the rain sum coefficient (-2.00), finally, underlines how precipitation strongly reduces ozone maxima.

By analyzing the effect that every month has on the median of weekly ozone (Figure A.6), we can observe a clear annual pattern with a peak in late spring

(April and May) and a minimum in autumn (October and November).
On the other side, by considering the values assumed by the hyperparameters spread and tail (Figure A.7), we can observe that:

- the first one has a mean of more or less 27.7 μg/m^3, indicating typical deviations of approximately $25 - 30$ μg/m^3 from the median and so a well identified moderate dispersion;

- the second parameter, the tail ξ, has a mean value of around 0.0003, really close to zero, telling us that we have an exponential Gumbel-type tail with no existence of heavy tails, implying that effectively extreme ozone events are rare.

Finally, we also decided to make out-of-sample predictions on the test set using this temporal bGEV model. In particular, once prepared test data, we made predictions by using the same likelihood, priors and hyperparameters used for the training set, without refitting spread, tail and RW2 precision, but applying all to the chosen test set. In this way, we can state that the model successfully captures the seasonal dynamics and overall level of weekly ozone concentrations with an increase from winter to summer, summary peaks and an autumnal decrease. However, extreme summer peaks tend to be slightly underestimated, suggesting that the tail behaviour may be somewhat too light or that additional heteroscedastic structure could improve the fit.

4.2 bGEV heteroskedastic model

The second model to analyze for our case is a bGEV heteroskedastic model, not based on including any temporal component, but on adding covariates on the spread parameter:

$$
\begin{aligned}
y_{i,t} \mid q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi \;&\overset{\text{ind}}{\sim}\; \text{bGEV}\big(q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi\big) \\
q_{\alpha,i,t} = \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta}, &\qquad s_{\beta,i,t} = \exp(\gamma_0 + \mathbf{x}_{i,t}^{\top}\boldsymbol{\gamma}), \\
\boldsymbol{\beta},\, \boldsymbol{\gamma},\, \xi \;&\sim\; \pi\big(\boldsymbol{\beta}, \boldsymbol{\gamma}, \xi\big) = \pi(\boldsymbol{\beta}) \cdot \pi(\boldsymbol{\gamma}) \cdot \pi(\xi), \\
\textit{with } \boldsymbol{\beta} &\sim \mathcal{N}\big(\mathbf{0},\, 1\big), \\
\gamma_0 &\sim \text{Gamma}(3, 3) \\
\boldsymbol{\gamma} &\sim \mathcal{N}\big(\mathbf{0},\, 1\big). \\
\xi &\sim \text{P3C prior}, \quad \xi \in [0,\, 0.5]
\end{aligned}
$$

In order to deal with this model, we used a procedure similar to the one we have applied before. In fact, we started by defining the priors for the spread and for the tail parameter, before computing all the hyperparameters of the regression coefficients affecting the spread and assembling all bGEV hyperparameters together. Later, we also fixed the median and the 50% interquartile range and we added the heteroskedastic part of the model by describing the spread not as a constant quantity, like in the previous case, but as linearly depending on the four meteorological covariates of the dataset. Then, we ended up this methodology by building INLA input and fitting the corresponding model using the "inla" function.

The resulting effects with this model on location can be interpreted in the following way (Figure A.8):

- the intercept of 99.13 μg/m^3 is slightly lower with respect to the intercept of the previous model and corresponds to a baseline weekly maximum ozone level under average conditions;
- the minimum temperature coefficient $(7, 92)$ implies that higher minimum temperatures increase ozone maxima, in contradiction with the result obtained for the first model;
- the maximum temperature is still the dominant driver of ozone maxima, but with smaller magnitude than before (now 28.7), with the extreme behaviour now explained by variance inflation rather than a shift in the median;
- the wind speed covariate now shows a stronger effect (0.87);
- the rain sum has a negative coefficient (-3.58), showing a stronger suppressing effect.

By considering, instead, the results for the spread and tail effects (Figure A.9), they can be described as follows:

- the baseline spread of 34.8 μg/m^3 is larger than in the previous model and with covariates-dependent variability represents an average level of spread across meteorological regimes;
- the maximum and minimum temperature coefficients are really small, implying that these two covariates don't have a significant effect on the spread;
- the wind speed is a strongly significant covariate for the spread, stabilizing ozone extremes by reducing variability and limiting extreme spikes with a coefficient of -0.166 in a log scale;

- the rain sum, instead, increases variability despite lowering the mean, showing a positive coefficient of 0.029;
- the tail , finally, assumes a small value of 0.0026 more or less, remaining light, but mildly heavier than in the previous case after accounting for the heteroskedasticity. This brings to the conclusion that constant-spread models tend to force excess variance into the tail.

In the end, we decided to make out-of-sample predictions of ozone maxima on the test set by using this second model. As before, we opted for considering the same structure already defined in the training set, with constant tail, same formula for the spread and no refitting of hyperparameters, with the aim to study if for a certain sensor the predicted data were close enough to the observed ones. We concluded that the heteroskedastic model captures the overall seasonal pattern, even if extreme summer peaks remain slightly underestimated, a consistent result considering the near-zero tail parameter.

4.3 Models comparison

Once introduced and developed two different models, we can compare these two ways of analysis using different possible criteria like the tail parameter ξ, the spread, the maximum temperature and the underestimation of the extremes. In particular, the temporal model focuses on the seasonal and annual point of view of the problem, showing a really strong effect of the covariate maximum temperature and a constant spread; the heteroskedastic model, instead, bases its analysis on the dependence of the spread on different covariates with a higher value of baseline spread, but a weaker effect of the maximum temperature.

Moreover, the PIT histogram is a really useful tool to evaluate if predicted probabilities match observed data for a certain model and it has the characteristic of being uniformly distributed in case of well-calibrated models. If we look at the PIT histograms of the two models (Figure A.10), we can observe that both representations show Inverted-U shape and shift to the right, meaning that both models present overdispersion with an under-estimation of high ozone extreme values.

4.4 Model with covariates on spread and temporal components

One of the possible next steps of analysis is to merge the two previously defined models to obtain one more complete model with temporal components in the formula of the location and some covariates influencing the computation of the spread, like in the following way:

$$
\begin{aligned}
y_{i,t} \mid q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi \; &\overset{\text{ind}}{\sim} \; \text{bGEV}\big(q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi\big) \\
q_{\alpha,i,t} &= \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta} + \beta_{year} year + f(month) \\
s_{\beta,i,t} &= \exp(\gamma_0 + \boldsymbol{\gamma_1}\text{windSpeed} + \boldsymbol{\gamma_2}\text{RainSum}), \\
\boldsymbol{\beta},\, \boldsymbol{\gamma},\, \xi \;&\sim\; \pi\big(\boldsymbol{\beta}, \boldsymbol{\gamma}, \xi\big) = \pi(\boldsymbol{\beta}) \cdot \pi(\boldsymbol{\gamma}) \cdot \pi(\xi), \\
with\ \boldsymbol{\beta} &\sim \mathcal{N}\big(\mathbf{0},\, 1\big), \\
\gamma_0 &\sim \text{Log-gamma}(3, 3) \\
\boldsymbol{\gamma} &\sim \mathcal{N}\big(\mathbf{0},\, 1\big). \\
\xi &\sim \text{P3C prior}, \quad \xi \in [0,\, 0.5]
\end{aligned}
$$

Repeating the identical procedure we have seen before, we computed the PC prior for the tail and the priors for the spread, for the regression coefficients affecting the spread and for the monthly seasonal effect. In a second moment, we set the bGEV model-specific quantile and mixing settings with the aim to generate the control list for the bGEV family, to build the INLA input, to define the model formula with all necessary components and then to fit the final model.
So, we obtained the following important results (Figure 8):

- the intercept of 106.92 μg/m^3 corresponds to the reference high-ozone quantile when meteorological covariates are at their centered values and seasonal effect is zero, with the value influenced by the temporal structure of the model and being really close to the one obtained for the first model;

- the year effect is synthesized by the coefficient -0.05 μg/m^3, implying no clear long term trend in extreme ozone concentrations over the study period;

- the minimum temperature has weak and statistically uncertain effects on location, with nighttime temperatures not strongly influencing extreme ozone levels;

- the maximum temperature always remains the dominant meteorological driver, with hot daytime conditions strongly amplifying ozone extremes;
- the wind speed has a strong negative effect on location, with increased wind speeds effectively reducing extreme conditions;
- the rain sum has a negative effect as well, meaning that precipitation significantly suppresses ozone extremes.

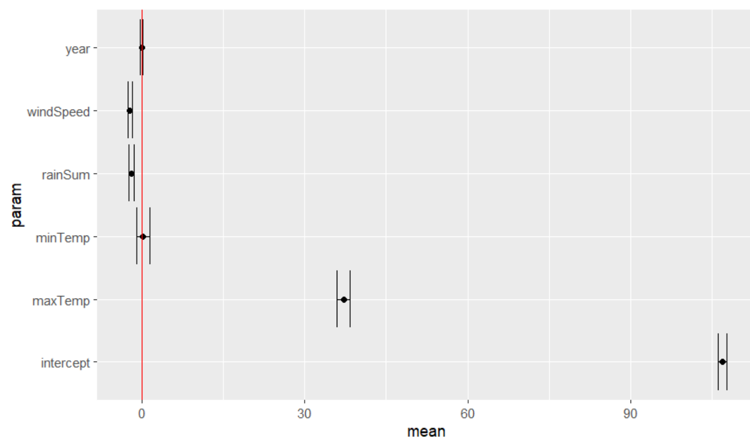


Figure 8: Parameters of hybrid model

If we concentrate, instead, on the posterior summary of the bGEV hyperparameters (Figure 9), we can notice that:

- the baseline spread, equal to 35.53 μg/m^3, represents the typical variability of extreme ozone concentrations under average meteorological conditions and permits us to state that the structural variability is really stable comparing this result with the previous ones;
- the wind speed has a strongly significant effect, stabilizing ozone extremes by reducing variability and limiting extreme spikes;
- also the rain sum is a significant covariate, slightly increasing variability despite lowering the mean;
- the tail ξ assumes a really small value, 0.0002 more or less, underlying again the presence of light tails and so no evidence of heavy-tailed extremes.

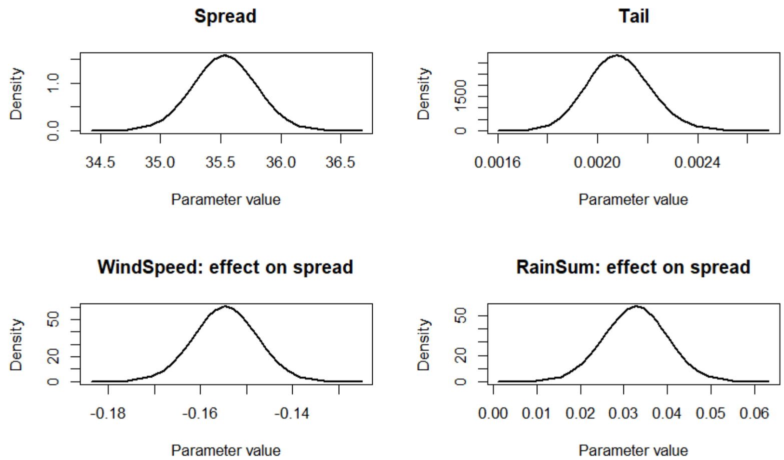


Figure 9: Marginal posterior densities of hyperparameters for hybrid model

Moreover, the PIT histogram of the merged model is very similar to that of the second model, displaying an inverted U-shape with a slight rightward shift (Figure 10). This pattern suggests residual miscalibration. The comparison between observed and predicted ozone levels, illustrated for sensor 5707 (Figure 11), shows that the predicted values reproduce the overall seasonal pattern. However, they frequently remain below the observed values, particularly during extreme summer peaks, indicating a systematic underestimation of high ozone concentrations.

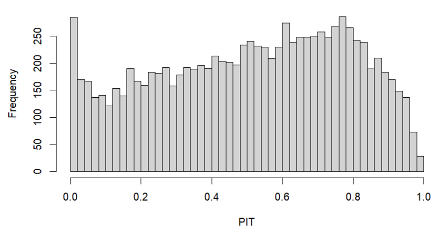


Figure 10: PIT histogram for the hybrid model

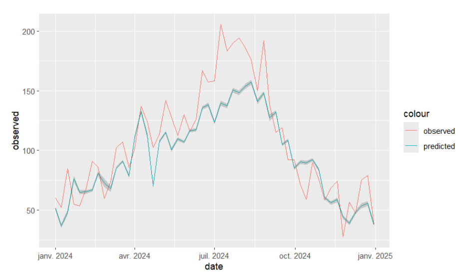


Figure 11: Observed vs predicted weekly maximum ozone (sensor 5707)

5 Stan implementation of the bGEV model

In order to validate the INLA-based inference and to assess the robustness of the Laplace approximation, we implemented an equivalent Bayesian bGEV model in `Stan`, performing full posterior sampling via Hamiltonian Monte Carlo.
The regression structure adopted in Stan is simpler than the previous hybrid formulation and focuses on a linear predictor for the location parameter. In particular, we define

$$q_i = X_i \boldsymbol{\beta}$$

Since the bGEV parametrization is defined in quantile space, a non-linear transformation is required to recover the classical GEV parameters. Therefore, the median parameter q_i, together with the spread s and the tail parameter ξ, is transformed into the classical location and scale parameters (μ_i, σ_i) through a deterministic mapping:

$$(\mu_i, \sigma_i) = f(q_i, s, \xi)$$

Conditional on these parameters, the observations are assumed to follow the blended GEV distribution:

$$y_i \sim \mathrm{bGEV}(\mu_i, \sigma_i, \xi)$$

5.1 Prior specification

The prior structure mirrors the one used in the INLA framework to ensure comparability between the two approaches. In particular:

$$\beta_j \sim \mathcal{N}(0, 1)$$

which represents weakly informative priors centered at zero for all regression coefficients.
For the spread parameter we adopt a log-normal prior,

$$s \sim \mathrm{LogNormal}(-0.14, 0.29)$$

ensuring positivity while being similar to Gamma(3,3) used in INLA.
Finally, the tail parameter is assigned a truncated normal prior,

$$\xi \sim \mathcal{N}(0, 0.2),\ 0 \leq \xi \leq 0.5$$

which regularizes the shape towards light tails while still allowing moderate positive tail behaviour like the P3C prior.

5.2 Posterior inference

Posterior inference in Stan is performed using Hamiltonian Monte Carlo (HMC), which provides asymptotically exact samples from the full posterior distribution. This contrasts with INLA, which relies on deterministic nested Laplace approximations.
To ensure a fair comparison, we used:

- the same weekly ozone maxima,
- the same three monitoring stations,
- identical covariates and preprocessing steps.

5.3 Comparison between Stan and INLA

The posterior means obtained from both methods are summarized in the following table.

Parameter	INLA Mean	Stan Mean
Intercept	10.78	9.357
minTemp	1.46	1.323
maxTemp	1.93	1.156
windSpeed	3.32	0.078
rainSum	-0.02	0.203
s	59.66	120.517
xi	0.0029	0.003

Overall, we observe a good agreement for most regression coefficients. The intercept and the effects of minimum and maximum temperature are of comparable magnitude, and the estimated tail parameter ξ remains extremely small in both frameworks, around 0.003, confirming the presence of light-tailed behaviour (Figure A.11).
However, a larger discrepancy appears for the spread parameter s, with Stan producing a higher posterior mean compared to INLA (Figure A.12). Differences are also visible in some meteorological effects, particularly wind speed and rain. These discrepancies are likely attributable to the different inference strategies: INLA relies on Laplace approximations around posterior modes, while Stan performs full posterior exploration via MCMC sampling.

Despite these differences in magnitude for certain parameters, the qualitative conclusions remain consistent. In both implementations:

- maximum temperature emerges as a key driver,
- the spread parameter captures substantial variability,
- the tail parameter remains close to zero.

Github Repository

`https://github.com/alebouvier/BS_project`

6 References

- Gabriel Escarela. Extreme value modeling for the analysis and prediction of time series of extreme tropospheric ozone levels: A case study. *Journal of the Air Waste Management Association*, 62(6):651–661, 2012. doi:10.1080/10962247.2012.665414.
- Edzer Pebesma and Benedikt Gräler. *Introduction to spatio-temporal variography*, 2025.
- Daniela Castro-Camilo, Raphaël Huser, and Håvard Rue. Practical strategiesfor generalized extreme value-based regression models for extremes. *Environmetrics*, 33(6):e2742, 2022. doi: https://doi.org/10.1002/env.2742. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/env.2742.

7 Appendix

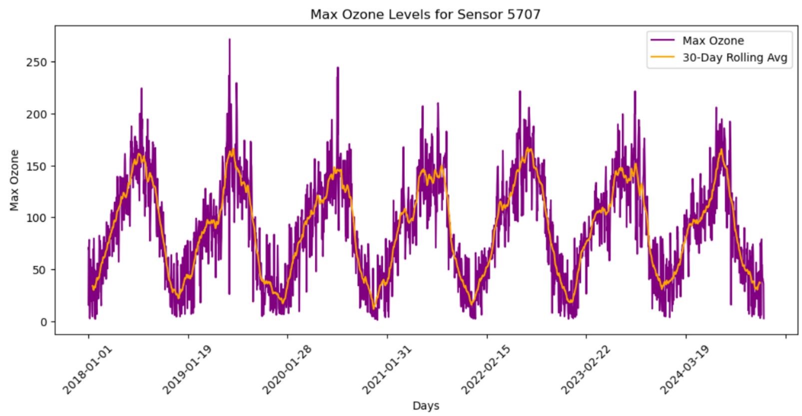


Figure A.1: Evolution of the daily ozone maximum for sensor 5707.

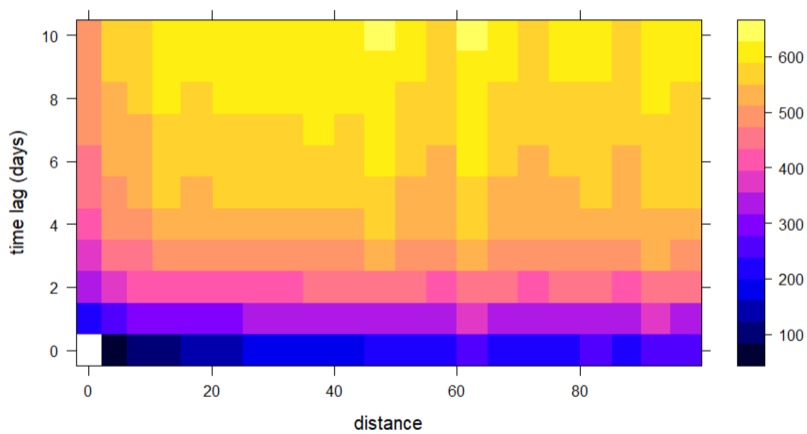


Figure A.2: Spatio temporal variogram.

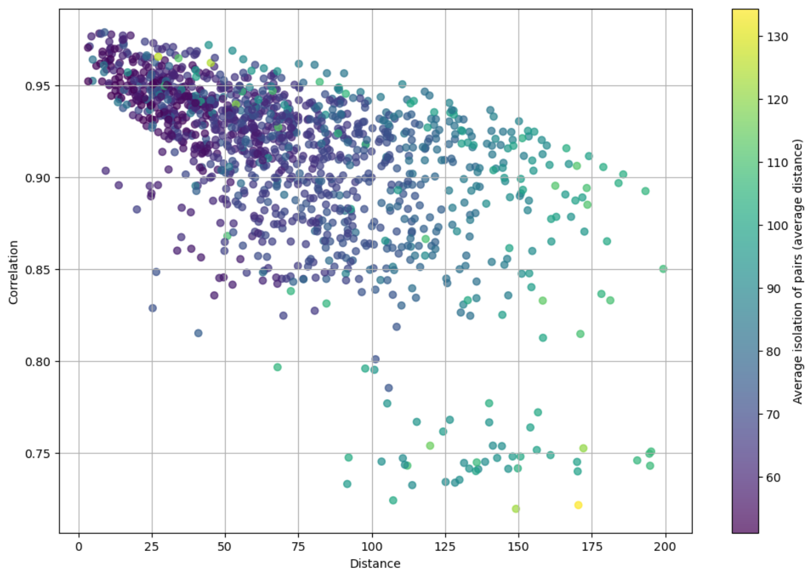
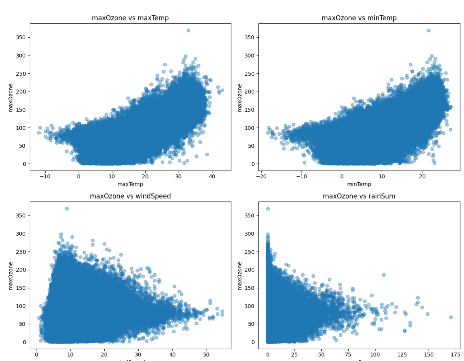
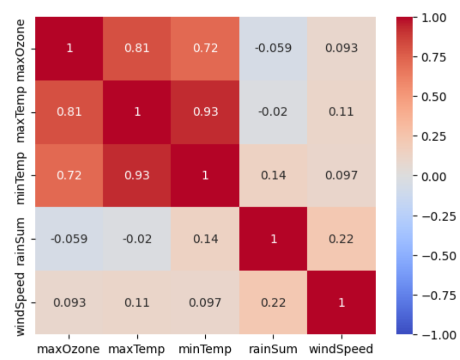


Figure A.3: Cross correlation between sensors.



(a) Scatterplot



(b) Correlation matrix between variables.

Figure A.4: Scatterplot and correlation matrix

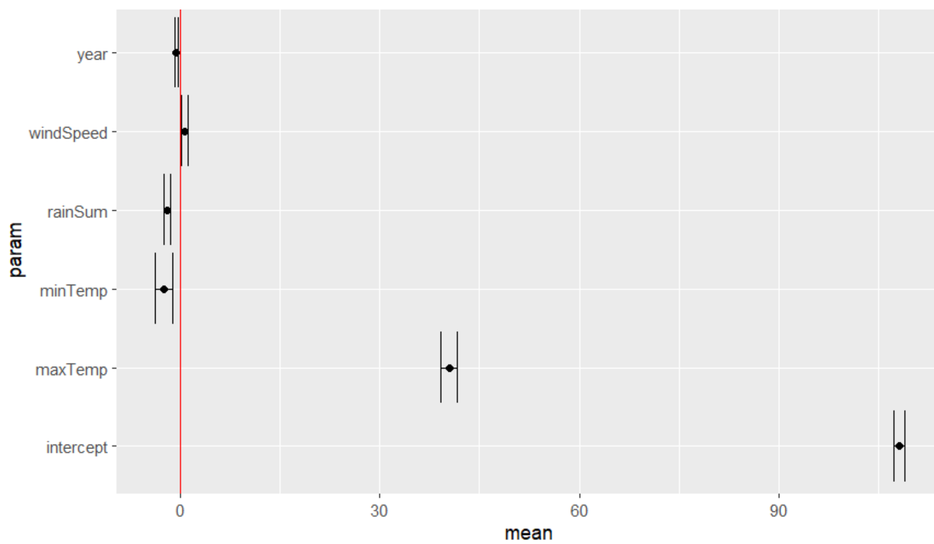


Figure A.5: Parameters of temporal model

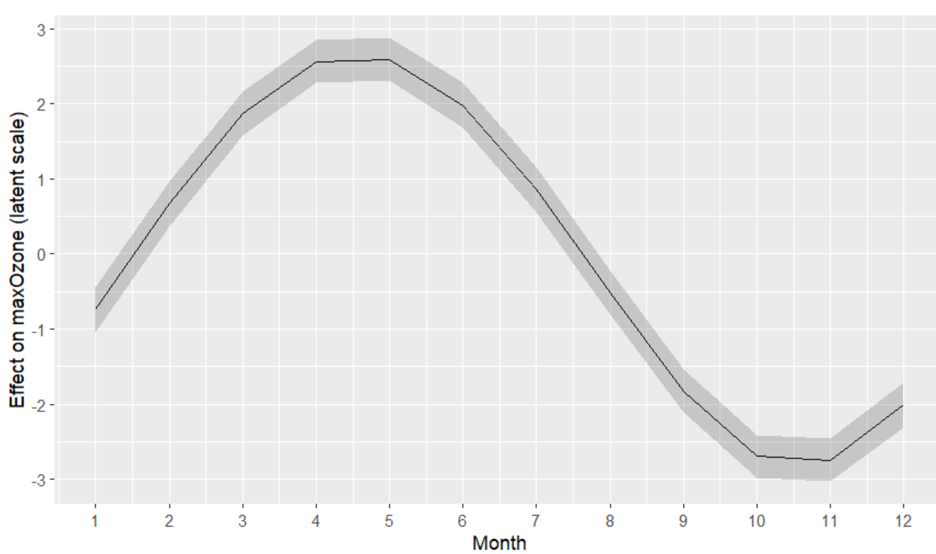


Figure A.6: Seasonal effect of month for temporal model

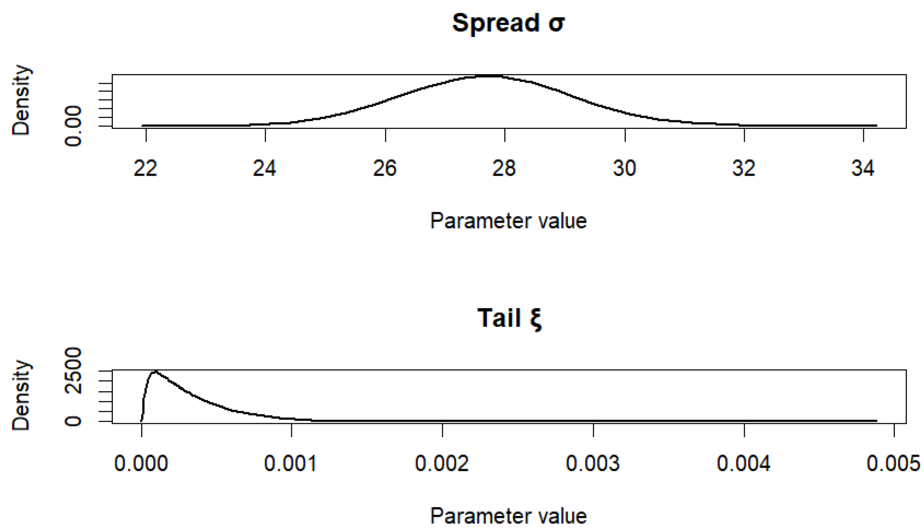


Figure A.7: Marginal posterior densities of hyperparameters for temporal model

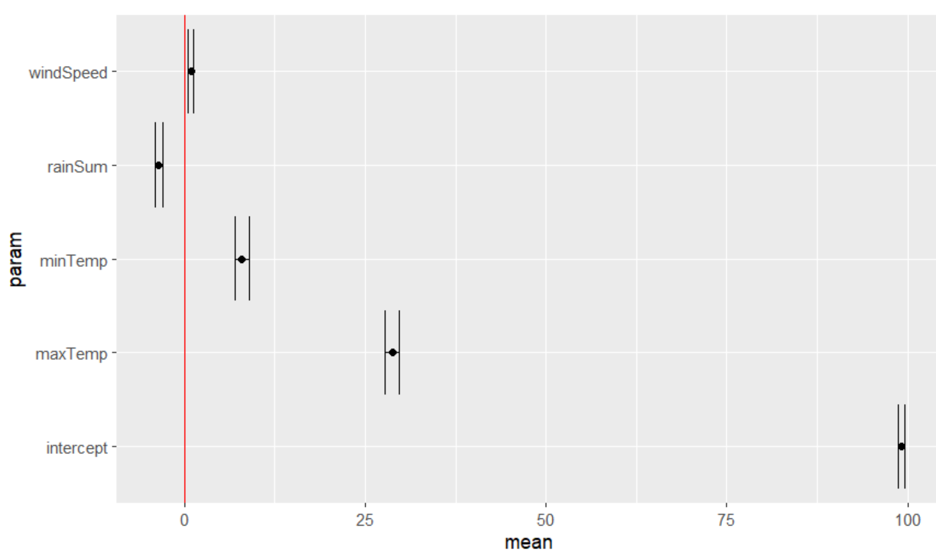


Figure A.8: Parameters of heteroscedastic model

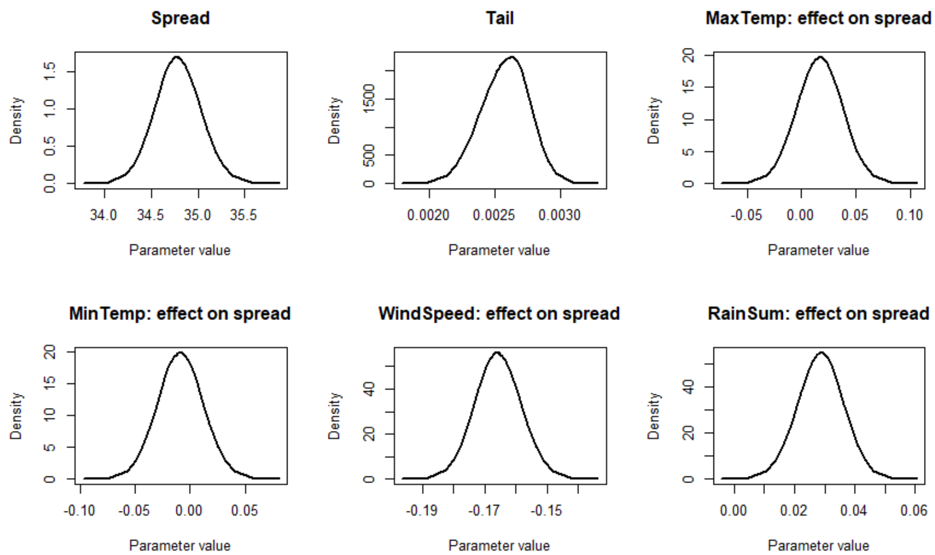


Figure A.9: Marginal posterior densities of hyperparameters for heteroscedastic model

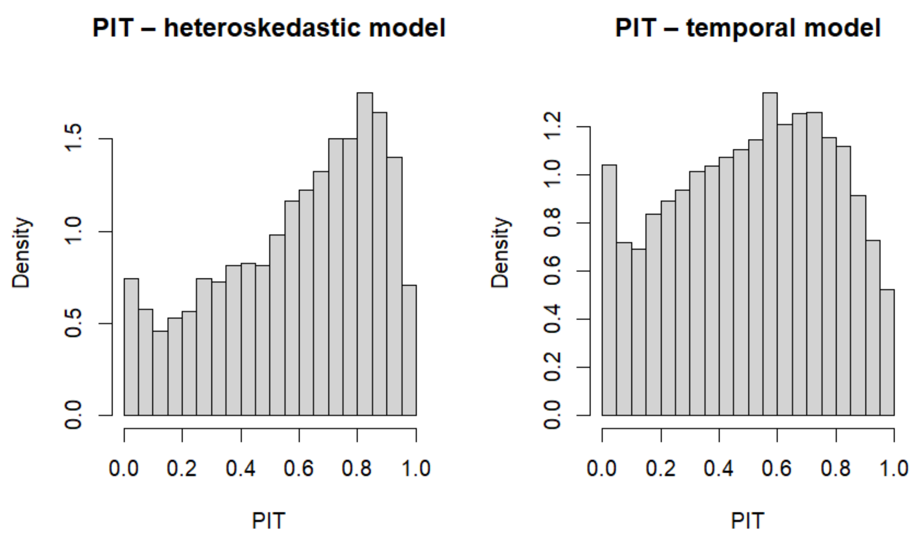


Figure A.10: PIT comparison

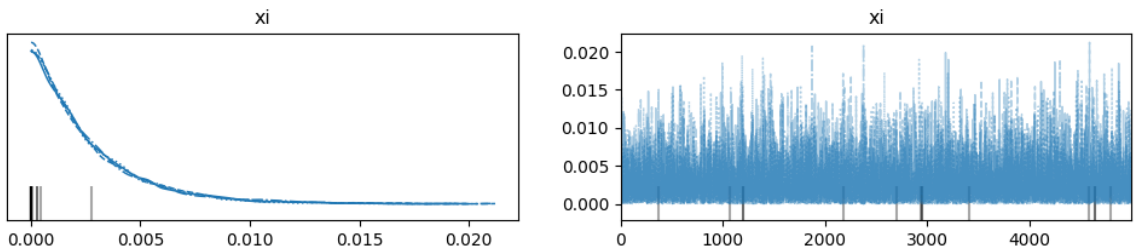


Figure A.11: posterior distribution of the tail parameter in STAN implementation

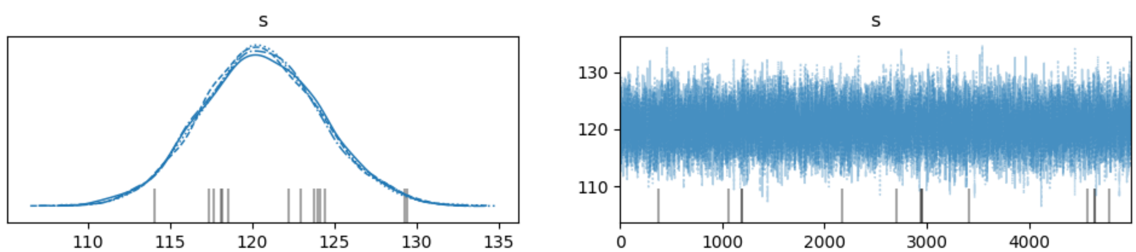


Figure A.12: posterior distribution of the spread parameter in STAN implementation